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Responsible for the structure of the five-stage pipeline and the data that flows through it.

Files and document sections

File / section	Content
`src/pipeline-model.js`	The four pipeline registers (IF/ID, ID/EX, EX/MEM, MEM/WB) and the no-control signal set.
`docs/M2-design.md §2–§3`	Architecture, file table, stage-level and datapath-level block diagrams.
`docs/M2-design.md §4–§5`	Pipeline-register fields and the control-signal table.
`src/pipeline.js (planned, M3)`	The clock-cycle engine.

Work allocated

- Architecture of the simulator as a static browser page and the role of each source file.
- Block diagrams of the pipeline: IF, ID, EX, MEM, WB with the four pipeline registers, and a datapath-level view showing where the control, hazard and forwarding units sit.
- Data model of the four pipeline registers: what each stage reads from its left register and writes to its right, plus the bubble marker for stalls and flushes.
- Control signals (RegWrite, ALUSrc, ALUOp, MemRead, MemWrite, MemtoReg, Branch, PCSrc) and their values for all 15 instructions.
- Rule that a bubble carries all-zero control signals so it can never write a register or memory.
- Design of the engine as a pure function step(state) → newState, which makes Step Back simple and the engine testable without a browser.